Canonical two-chunk regional simulations (en)

Canonical two-chunk regional simulations — ULVZ project guide

> **Read this page first.** This project-local guide covers only the accepted canonical mode: NCHUNKS=2, two adjacent 90°×90° faces, AB central, AC joined on the supported local side, a single center/gamma orientation, and the accepted patch. It is not an upstream SPECFEM3D_GLOBE promise. canonical_90deg_fixture_ready=true; general_two_chunk_mode_classification=B.

> **Do not extrapolate.** Non-90° or unequal widths, other attachment sides, nonadjacent faces, independent orientations, arbitrary two-chunk topologies, and three-/general multi-chunk modes are unverified. The planner is read-only: it does not replace mesher, Stacey, or waveform acceptance. Kim/Song exact reconstruction still requires author inputs.

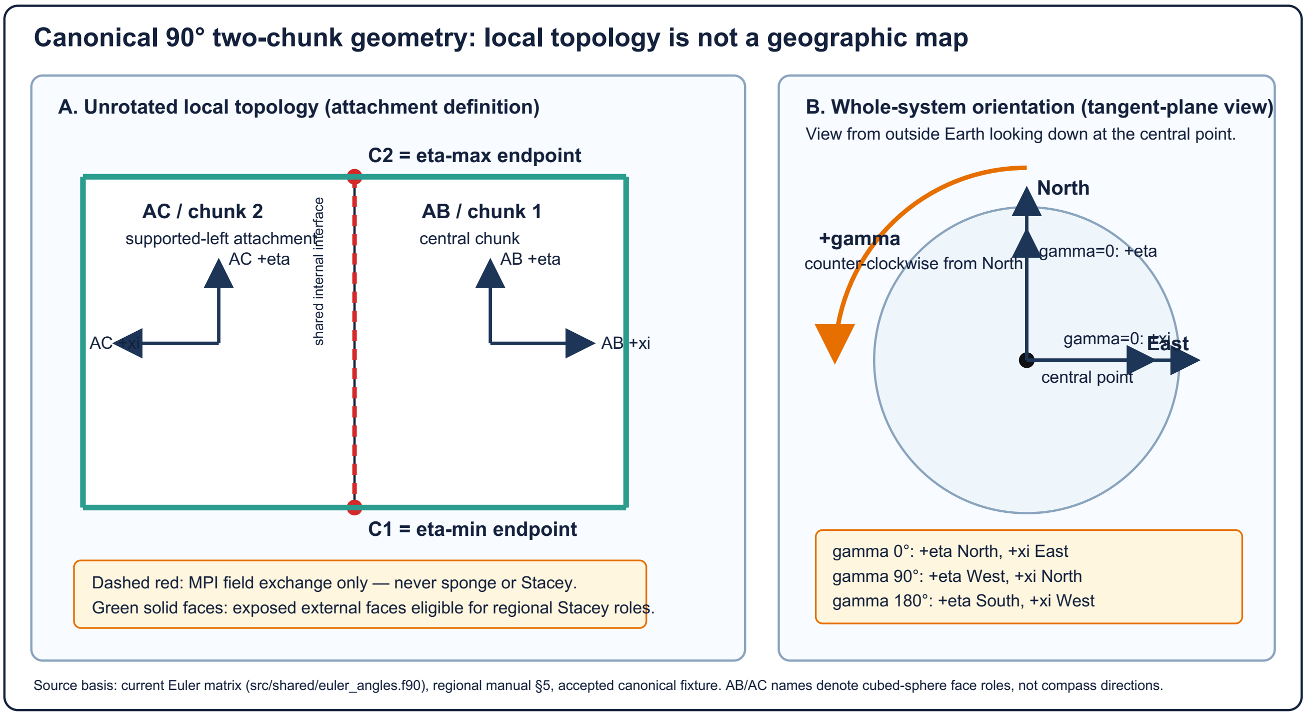
1. What this mode is, and the words used below

A regional simulation computes only selected cubed-sphere faces. One chunk is the usual regional case; six chunks form the full sphere. This two-chunk mode is useful only when the source, receivers, and target paths do not fit one 90° face but do fit these two adjacent faces. It is not a free-form mesh.

Term	Plain meaning and location	Why a user must care / visible failure
AB / AC	The two faces used here. AB is central chunk 1; AC is attached chunk 2.	They identify the owner of a source or receiver. Swapping them changes supported topology.
AB–AC interface	The common interior face: AB xi-min meets AC xi-max.	It exchanges wavefield data; it is not an exterior wall and must never absorb.
C1 / C2	The two interface ends: C1 is eta-min; C2 is eta-max. Eta is face-local, not geographic south/north.	Endpoint errors cause missing or wrongly connected corner data and may vary by MPI layout.
MPI rank / rank zero	A rank is one MPI process; rank 0 is a valid process.	Rank 0 must not be treated as an unused slot.
buffer / two-member path	A buffer carries endpoint data between the two ranks meeting at C1 or C2.	Each endpoint needs exactly two reciprocal members in this mode.
three-member / BC path	Historical general-face corner logic included a BC participant not in this two-face domain.	It can create a false connection or conceal a missing endpoint.
Stacey face / sponge	Stacey absorbs on an exposed mesh face. A global sponge is a high-attenuation region in a six-chunk global model.	Neither may be put on AB–AC.

Implementation background. Before the accepted patch, C1 had no eta-min two-member record and C2 retained a BC/rank-zero three-member path. The patch creates two endpoint records, uses INVALID_RANK only for unused slots, and segments the xi-constant interface with NPROC_ETA.

2. Geometry: local attachment is not a map direction



The left panel is a **local topology drawing**, not a map. "Supported-left" means that AC connects to AB's designated local edge: AB xi-min joins AC xi-max. Once center latitude/longitude and gamma rotate the complete system, AC may appear west, east, north, or south on a map. Say "attached to the specified local edge", not "geographically left".

The dashed AB–AC line is internal. C1 is eta-min and C2 eta-max. Remaining lateral faces are exposed. An endpoint can share a *node* with an external Stacey face without making the internal face absorbing: face roles are different. Accepted topology evidence found zero internal AB–AC Stacey faces at shallow, mid-mantle, and CMB-near depths.

Current source stops when NCHUNKS > 1 and ANGULAR_WIDTH_XI_IN_DEGREES is not 90. It permits a non-90 eta width only for exactly two chunks, but this project has not validated that case: set both widths to 90.

3. Center coordinates and GAMMA_ROTATION_AZIMUTH

CENTER_LATITUDE_IN_DEGREES and CENTER_LONGITUDE_IN_DEGREES place AB's center. GAMMA_ROTATION_AZIMUTH rotates the **complete AB+AC system**; AC has no independent orientation. The official regional manual says gamma is measured counter-clockwise from due north. Current euler_angles.f90 and the project planner use the same sign.

Viewing convention: look from outside the Earth toward the selected center, with north up and east right in the tangent plane. Positive gamma is counter-clockwise. With ELLIPTICITY=.false. and center=(0°,0°):

Gamma	+eta	+xi	Meaning
0°	north	east	reference orientation
90°	west	north	complete system rotated counter-clockwise
180°	south	west	half turn

With ellipticity enabled the source first converts geographic latitude to geocentric colatitude; this does not change gamma's sign. Audit a source and all stations after every center/gamma change. Never rotate AC separately.

Determine AC's actual geographic position after setting parameters

There is no AC latitude, longitude, or gamma parameter. For canonical geometry, the four inputs `NCHUNKS=2`, both 90° widths, AB center latitude/longitude, and gamma determine every AB and AC boundary point uniquely: SPECFEM first defines the fixed local AB--AC pair, then applies the same Euler transform to the whole pair. AC is therefore not a second region that the user places.

For a selected center/lon/gamma, use `two_chunk_planner plan` with the same minimum and maximum for each search range. Its `map.png` shows the transformed AB/AC outline and interface, while `globe.png` shows their closed spherical relationship; `geometry_audit.json` and `candidates.json` report source/station chunk labels, local xi/eta, interface, C1/C2, and exposed-boundary margins. This is the operational way to decide whether the intended geographic region falls in AB, AC, the shared interface, or outside. It is a pre-mesh geometric check, not a replacement for later mesh/Stacey/waveform validation.

At a geographic pole, longitude and gamma are not a unique parameterization. The planner deliberately normalizes a polar center to longitude 0° and adjusts gamma to preserve the same physical AB+AC orientation. Compare the plotted outline, not only the three printed numbers, when reviewing a polar case.

4. Install the accepted patch safely

The package is patches/specfem3d_globe/two_chunk_endpoints/. Its JSON manifest is the hash authority. It was validated against nested revision 9c312cb2c991b47484a7f302775f4f01ed9470f8. Baseline target SHA-256 is fd4137713e55e14ec664a9d55487b64c2b9bf73499c1f82780f1f5a6e63b088f; applied target SHA-256 must be 8c64f1d1d415ec6c0792f06474dafcffcc698da6ee03ecd21bfd4fdc90b64857.

Minimum installation

Run from the project root. Stop on a hash or context mismatch.

```
PATCH="$(pwd)/patches/specfem3d_globe/two_chunk_endpoints/specfem3d_globe_two_chunk_endpoints.patch"
patches/specfem3d_globe/two_chunk_endpoints/verify_patch.sh specfem3d_globe
git -C specfem3d_globe apply --check $PATCH
git -C specfem3d_globe apply $PATCH
sha256sum specfem3d_globe/src/meshfem3D/create_chunk_buffers.f90
git -C specfem3d_globe apply --reverse --check $PATCH
```

The verifier must report baseline hash, single-file scope, no DEBUG change, and a clean apply check. If the target already has the candidate hash, the reverse dry-run identifies it as applied; do not apply twice. To reverse intentionally: run git -C specfem3d_globe apply --reverse $PATCH, then rebuild from clean objects.

Strict traceability and manual replacement

Record nested revision, baseline hash, patch SHA-256 4496cea542d26f38575ec1fa9ae28635ec2a201958eb898702331c7db5fe4a60, forward check, applied hash, and reverse check. Prefer git apply because it checks context and leaves an auditable diff. Do **not** normally replace the complete create_chunk_buffers.f90 file. If site policy requires it, preserve the original, verify the exact baseline hash, compare the patch diff, verify the candidate hash, and retain a tested restore path. Copying a file is not two-chunk runtime acceptance. This patch does not cleanly apply to v8.0.0.

5. Inputs, external absorption, source, and stations

Start from current specfem3d_globe/DATA/Par_file. The teaching copy is cases/two_chunk_canonical_90deg/DATA/Par_file.

Setting	Canonical two-chunk instruction
NCHUNKS	Set 2. Total ranks = 2*NPROC_XI*NPROC_ETA. Accepted fixtures: 2, 8, 12 ranks.
ANGULAR_WIDTH_XI_IN_DEGREES, ANGULAR_WIDTH_ETA_IN_DEGREES	Set both 90.d0.
CENTER_LATITUDE_IN_DEGREES, CENTER_LONGITUDE_IN_DEGREES, GAMMA_ROTATION_AZIMUTH	Place/rotate whole domain; use a fixed-geometry planner check after every change.
NEX_XI, NEX_ETA, NPROC_XI, NPROC_ETA	NEX=96 and 2×2/chunk are the accepted teaching fixture, not generic cheap production accuracy. Keep source divisibility constraints.
MODEL, OCEANS, ELLIPTICITY, TOPOGRAPHY, GRAVITY, ROTATION, ATTENUATION	Choose physical assumptions deliberately. Teaching switches are not a scientific prescription.
RECORD_LENGTH_IN_MINUTES, NTSTEP_BETWEEN_OUTPUT_SEISMOS, NTSTEP_BETWEEN_OUTPUT_SAMPLE	Select duration/cadence from a target signal and documented external-return assessment.
ABSORBING_CONDITIONS	Set .true. for canonical two chunks. Mesher builds Stacey arrays only for exposed faces.
ABSORB_USING_GLOBAL_SPONGE, SPONGE_LATITUDE_IN_DEGREES, SPONGE_LONGITUDE_IN_DEGREES, SPONGE_RADIUS_IN_DEGREES	Set sponge flag .false. Current source stops if it is true with NCHUNKS=2.
REGIONAL_MESH_CUTOFF, REGIONAL_MESH_CUTOFF_DEPTH, REGIONAL_MESH_ADD_2ND_DOUBLING	Optional radial controls; retain allowed current-template values and recheck mesh/Jacobians.
LOCAL_PATH, LOCAL_TMP_PATH, SAVE_MESH_FILES, OUTPUT_SEISMOS_*	Plan new-run database, diagnostic, and waveform storage.

What actually absorbs in two chunks

Global sponge is a six-chunk global-model option. read_parameter_file.F90 reads SPONGE_* only when its flag is true, while read_compute_parameters.f90 stops any non-six-chunk request with "Please set NCHUNKS to 6 in Par_file to use ABSORB_USING_GLOBAL_SPONGE". Its Qmu modification is reached through the ATTENUATION model path; it is not a local side-face condition.

For NCHUNKS=2, create_regions_mesh.F90 invokes Stacey setup when ABSORBING_CONDITIONS=.true. get_absorb.f90 includes AC xi-min, AB xi-max, and exposed eta faces; it excludes AB xi-min and AC xi-max, the shared interface. A radial bottom face is added only where the source logic calls for it (outer core or regional cut-off). Therefore **AB–AC is never sponge, Stacey, or any absorbing face**. If a result says otherwise, stop for a face-role audit. The exact parameter guards and call path are documented in the absorbing-boundary source audit.

CMTSOLUTION contains PDE, event name, time shift, half duration, latitude, longitude, depth, and Mrr Mtt Mpp Mrt Mrp Mtp. STATIONS records are station network latitude longitude elevation burial. Sources/stations may be in either face, but avoid exact interface, endpoint, element face/edge, or exterior-face placement. Do not prescribe a fixed angular safety distance: estimate target arrival and earliest external return.

6. Custom simulation workflow and teaching fixture

Do not mix these two workflows. The first is for **your own prepared input**; the second only checks or runs the project's fixed teaching case. Neither is formal patch acceptance. Every writing command uses a new output directory.

A. Plan and run your own canonical configuration

1. Check source and patch (section 4). Prepare your own `DATA/Par_file`, `CMTSOLUTION`, and `STATIONS`; set two 90° widths, `ABSORBING_CONDITIONS=.true.`, and `ABSORB_USING_GLOBAL_SPONGE=.false.`.
2. If geometry is not yet chosen, use the package planner guide for a bounded center/lon/gamma search. Once a candidate is chosen, use the fixed-range command below to display and classify that exact geometry. Replace all values marked for replacement; the output directory must not exist.

```
# Set this to the project-managed interpreter, for example /path/to/ulvz-specfem/bin/python.
export ULVZ_PYTHON=/path/to/ulvz-specfem/bin/python
SPECFEM_ROOT="$(pwd)/specfem3d_globe"
INPUT_DATA="/absolute/path/to/YOUR_CASE/DATA"
CENTER_LAT=20.0
CENTER_LON=160.0
GAMMA=30.0
ANALYSIS_END_S=1900
PLAN_DIR="results/two_chunk_geometry_$(date -u +%Y%m%dT%H%M%SZ)"
PYTHONPATH=packages/two_chunk_planner/src $ULVZ_PYTHON -m two_chunk_planner plan \
  --cmtsolution "$INPUT_DATA/CMTSOLUTION" --stations "$INPUT_DATA/STATIONS" \
  --par-file "$INPUT_DATA/Par_file" --target-energy-end-s "$ANALYSIS_END_S" \
  --latitude-range "$CENTER_LAT,$CENTER_LAT" --longitude-range "$CENTER_LON,$CENTER_LON" \
  --gamma-range "$GAMMA,$GAMMA" --output "$PLAN_DIR"
```

3. Open `PLAN_DIR/map.png` and `PLAN_DIR/globe.png` to see the actual geographic and closed-sphere AB/AC boundaries. Read the chosen candidate's point classifications before continuing. If no feasible candidate is returned, or a required point is outside/on an endpoint/external boundary, stop and change geometry. Review—not blindly copy—`recommended_Par_file.inc`, then transfer its center/lon/gamma and any deliberately selected mesh settings into your own `Par_file`.
4. Run your own input only after the preceding review. `NPROCS=8` is the validated 2×2-per-chunk example; change it only to a compatible layout.

```
NPROCS=8
SPECFEM_ROOT="$(pwd)/specfem3d_globe"
INPUT_DATA="/absolute/path/to/YOUR_CASE/DATA"
RUN_DIR="results/my_two_chunk_run_$(date -u +%Y%m%dT%H%M%SZ)"
test ! -e "$RUN_DIR" || { echo "refusing to overwrite $RUN_DIR" >&2; exit 2; }
mkdir -p "$RUN_DIR/DATA" "$RUN_DIR/DATABASES_MPI" "$RUN_DIR/OUTPUT_FILES"
cp -R "$INPUT_DATA/." "$RUN_DIR/DATA/"
(cd "$RUN_DIR" && mpirun -np "$NPROCS" "$SPECFEM_ROOT/bin/xmeshfem3D")
(cd "$RUN_DIR" && mpirun -np "$NPROCS" "$SPECFEM_ROOT/bin/xspecfem3D")
```

`xmeshfem3D` produces the mesh and `DATABASES_MPI`; `xspecfem3D` reads those databases. This source tree has no separate `xgenerate_databases` command.

B. Check or run the fixed teaching fixture

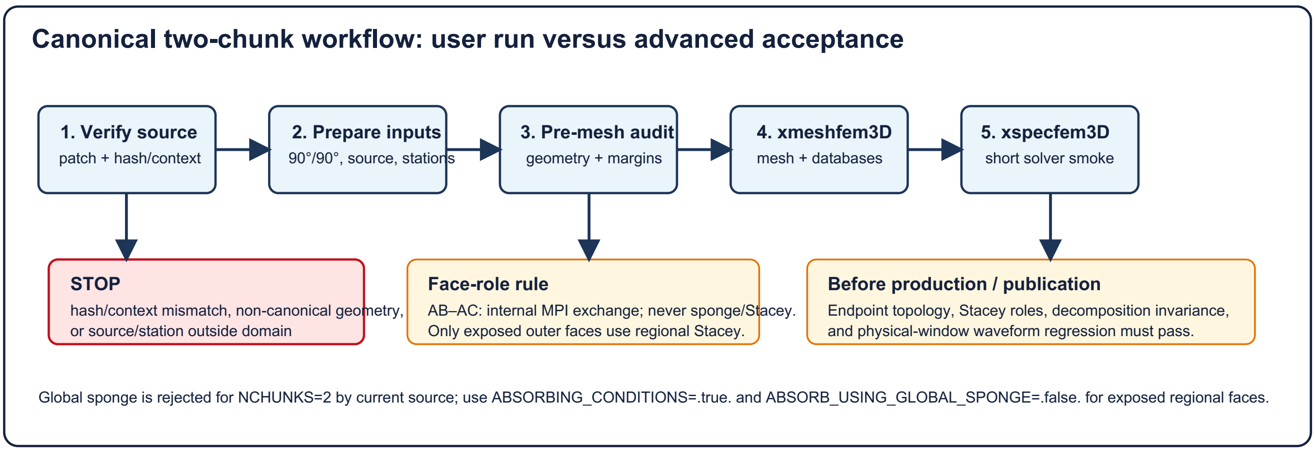
`audit_geometry.py` is intentionally restricted to `cases/two_chunk_canonical_90deg/DATA` (center 90°, longitude 90°, gamma 0°). `run_canonical.sh` copies that same teaching `DATA/`; it is not a reviewer and must not be used to launch your own inputs. Its dry-run is only a command preview for the teaching fixture.

```
SPECFEM_ROOT="$(pwd)/specfem3d_globe"
$ULVZ_PYTHON cases/two_chunk_canonical_90deg/audit_geometry.py --validate-only --specfem-root "$SPECFEM_ROOT"
bash cases/two_chunk_canonical_90deg/run_canonical.sh --specfem-root "$SPECFEM_ROOT" \
  --run-dir "results/two_chunk_canonical_run_$(date -u +%Y%m%dT%H%M%SZ)" --dry-run
```

Expected: the teaching audit reports status pass and total MPI ranks 8; its dry-run prints the 8-rank `xmeshfem3D` then `xspecfem3D` commands without creating the run directory. Remove `--dry-run` only to reproduce that teaching fixture, not to run a custom study.

7. Full production or release acceptance

Keep these checks separate from first use.



The diagram's pre-mesh step is the custom-input planner workflow in section 6A. The teaching audit and teaching runner in section 6B are separate fixture tools.

Level	Required evidence	Stop condition
Must	Patch context; canonical widths; in-domain points; finite mesher/solver output	mismatch, non-90° geometry, outside point, or mesher failure
Changed patch/source	C1/C2 reciprocal two-member paths; no INVALID_RANK member; zero internal AB–AC Stacey faces	missing endpoint, BC/three-member path, or internal absorbing face
Decomposition claim	Equivalent 2/8/12 physical fingerprints, connectivity, materials, buffers, face roles	unexplained rank difference
Waveform claim	Valid fixed physical window; no shift; no NaN/Inf; NRMS and relative energy <=1e-5	endpoint anomaly, invalid window, arrival shift, threshold failure
Recommended science	Target-arrival and external-return assessment; margins/path coverage	boundary contamination risk
Developer evidence	One-/six-chunk regression and controlled reversed reproduction	no broad causal claim without them

Accepted v3 achieved maximum NRMS 2.92e-6 and relative energy difference 2.86e-6 at 2/8/12 ranks. The earlier v2 [0,25] s window was before an approximately 52 s first arrival; it was invalid, not a reason to loosen gates.

8. Resolution and MPI choices

NEX_XI/NEX_ETA are lateral spectral-element counts; GLL points lie inside elements. Radial layers/doubling, model complexity, source bandwidth, timestep, and output cadence also matter. NEX=96 is an accepted canonical geometry fixture, not automatically a low-cost smoke mesh or production prescription.

Per chunk, decomposition is NPROC_XI × NPROC_ETA; total ranks are 2*NPROC_XI*NPROC_ETA. Accepted fixtures are 1×1/chunk=2, 2×2/chunk=8, and 2×3/chunk=12. Other compatible layouts are not project-validated. If mesher stops, choose compatible NEX/ranks; never bypass the source checks.

9. What each audit actually checks

Tool or record	Input	Output / pass meaning	It does not prove
verify_patch.sh	worktree and patch	source SHA, one-file scope, no DEBUG, clean apply context	topology or waveform
audit_geometry.py	fixed teaching DATA/ only	static teaching-fixture input/hash/membership check	arbitrary user center/gamma, databases, Jacobians, Stacey, waveforms
two_chunk_planner plan	user source/stations, candidate geometry, optional path/window	deterministic center/gamma candidates; actual AB/AC map and point classification	mesher, solver, or production-safe return time
stacey.bin analysis	generated databases	face roles; internal AB–AC count must be zero	science window
acceptance reports	controlled diagnostic runs	ownership, reciprocity, fingerprints, fixed-window metrics	general two-chunk support

There is no single “run audit tool”. Use the chain: source integrity → geometry → mesh/database face roles → solver/waveforms.

10. One-page checklist

- [ ] Patch hash/context and git apply --check agree.
- [ ] NCHUNKS=2; both widths 90; AB central and AC on specified local edge.
- [ ] ABSORBING_CONDITIONS=.true.; ABSORB_USING_GLOBAL_SPONGE=.false.
- [ ] Total ranks = 2*NPROC_XI*NPROC_ETA; NEX compatible.
- [ ] Source/stations are in-domain and not exactly on interface/C1/C2/faces.
- [ ] Window contains target arrival and documented return risk; it is not a universal window.
- [ ] Mesher completes with intended dimensions and acceptable Jacobians.
- [ ] New DATABASES_MPI/stacey.bin exist; required face-role audit finds zero internal AB–AC Stacey faces.
- [ ] Solver traces are finite and physical.
- [ ] Any symmetry claim has fingerprints and unshifted fixed-window metrics.

11. Troubleshooting

Symptom	Likely cause	Check	Safe response
“Please set NCHUNKS to 6 ... GLOBAL_SPONGE”	sponge enabled in two chunks	ABSORB_USING_GLOBAL_SPONGE	set .false.; use external Stacey
Claimed Stacey interface	node/face confusion or defect	face-role record	require zero AB–AC **face** records; stop if nonzero
Patch verifier fails	unmatched or already-applied source	baseline/candidate SHA, reverse check	do not force/overwrite; revalidate version
Mesher rejects ranks/NEX	incompatible decomposition	Par_file and stop text	choose compatible layout
Point outside	center/gamma or coordinates wrong	planner for custom input; teaching audit for fixture	move point or complete system, then rerun the applicable check
Zero/invalid trace	excluded receiver/input/solver failure	receiver list/log	correct inputs in a new run
Poor pre-arrival comparison	invalid window	travel-time assessment	redesign window; never shift traces
Rank disagreement	settings/mesh defect	fingerprint/raw traces	reproduce canonical inputs and stop if unexplained

12. Teaching case, glossary, and evidence

cases/two_chunk_canonical_90deg/ is a teaching fixture, not Kim/Song Hawai‘i input or a production model. It contains NEX=96, 2×2/chunk, an isotropic 50 km source, stations in both faces, audit_geometry.py, figure source, and the deliberate runner. Static audit should report seven in-domain points and eight total ranks. Its audit and runner do not audit or launch a custom input case.

Glossary. An external boundary is exposed and may be Stacey. The internal interface exchanges wavefield and can never be sponge/Stacey. C1/C2 are interface endpoints; shared nodes do not merge face roles. A fingerprint is a physical mesh summary, not a count of MPI files.

Evidence. Official regional behavior is in specfem3d_globe/doc/USER_MANUAL/05_regional_simulations.tex. Current geometry is in euler_angles.f90, write_profile.f90, and read_compute_parameters.f90. Absorption is in read_parameter_file.F90, create_regions_mesh.F90, get_absorb.f90, get_model.F90, and meshfem3D_models.F90. Patch hashes are in patches/specfem3d_globe/two_chunk_endpoints/. Accepted topology and waveform closure are in docs/two_chunk_corner_topology_acceptance.md and docs/two_chunk_waveform_symmetry_closure.md. The documentation-change record is the revision report.